Asadullah Shafique

Agentic AI Engineer | Multi-Agent Systems Architect | Enterprise AI Strategist

Bridging AI potential with financial services reality through orchestration, compliance, and scalable adoption.

Karachi, Pakistan +92-321-3771445 asadullahshafique@hotmail.com
github.com/asadullah48 LinkedIn Portfolio Medium

Professional Summary

Agentic AI Engineer building multi-agent systems for regulated industries, backed by **25+ years** of operating a manufacturing business. Most agentic pilots in financial services die at the compliance review — not because the models are weak, but because a system that cannot show *why* it reached a conclusion cannot be signed off by a second line of defence. My work makes that reviewable: typed orchestration with an auditable route, a written constitution with a deterministic layer that needs no model, and evaluation that reads the execution trace rather than trusting the prose.

Founder of Texcot Embroidery Sourcing House and creator of the Textile ERP Platform, targeting fabric mills, CMT units and garment exporters across Faisalabad, Sialkot, Gujranwala, Karachi and Lahore — so the domain intelligence is built from inside an industry rather than modelled from outside one. Completed **6 consecutive Panaversity Hackathons** (Bronze → Silver → Silver → Gold → Platinum, then Agent Factory), each build carrying the last one forward.

Agentic AI Mastery

Twelve competencies, four categories. Each cites the path that demonstrates it.

AGENTIC ORCHESTRATION

تنسيق الوكلاء

- Engineering Multi-Agent Ecosystem Design (MAS)
  `backend/orchestration/`
- Designing Graph-Based Agent Workflows
  `LangGraph · Plan-Act-Verify`
- Implementing Agentic Coding & SDLC Standardization
  `CLAUDE.md · spec-first`

CONNECTIVITY & PROTOCOLS

الاتصال والبروتوكولات

- Engineering MCP Interoperability Layers
  `/mcp/server · FastMCP`
- Standardizing Read-Only Agent Tool Contracts
  `6 tools, read-only`
- Architecting Provider-Agnostic Fallback Ladders
  `SDK → LangGraph → static`

ENTERPRISE & SCALE

المؤسسات والتوسّع

- Engineering Domain-Specific Intelligence (Finance)
  `FinAgent-Nexus · 94 tests`
- Architecting Domain-Specific ERP Intelligence (Textile)
  `CMT · live`
- Implementing Cloud-Native Scale-Out
  `k8s/ · 11 manifests`

RELIABILITY & SAFETY

الموثوقية والأمان

- Implementing Deterministic Guardrails
  `constitution/ · 4-of-4 blocked`
- Designing Feedback-Driven Evaluation Loops
  `evals/ · 3 suites`
- Engineering Context Integrity & Trace Auditability
  `orchestration/context.py`

Technical Skills

LANGUAGES

TypeScript | JavaScript | Python | HTML/CSS

FRONTEND

Next.js 15 | React 19 | Tailwind CSS v4 | shadcn/ui

BACKEND

FastAPI | SQLAlchemy ORM | PostgreSQL | REST APIs

AI & AGENTS

OpenAI Agents SDK | MCP Servers | LangGraph (LangChain ecosystem) | Constitutional AI | RAG | Qdrant | OpenAI API

CLOUD & DEVOPS

Kubernetes | Docker | Dapr | Kafka | Helm | GitHub Actions | Vercel | Koyeb | Minikube | Oracle Cloud (OCI)

OBSERVABILITY

Prometheus | Grafana | Jaeger | Loki

PLATFORMS & TOOLS

Git | Windows 11 | WSL2

Hackathon Achievements

Panaversity Hackathon Series: 6 consecutive builds · Bronze → Platinum, then Agent Factory

Completed 2025

Agent Factory (H5)

General Agent (Claude Code) manufactures Custom Agents (OpenAI Agents SDK). SKILL.md monetization. Digital FTE deployed on Kubernetes + Dapr. 117-slide deck.

Claude Code | OpenAI SDK | MCP | SKILL.md | Dapr | K8s

Platinum 2025

Cloud-Native Deployment (H4)

Full K8s stack: Kubernetes + Docker + Dapr + Kafka + Prometheus/Grafana/Jaeger + Helm + GitHub Actions CI/CD.

K8s | Docker | Dapr | Kafka | Helm | Prometheus

Gold 2025

Advanced Todo (H3)

149 tests passing. Recurring todos, team collaboration, AI suggestions, triple-layer Constitutional AI (7 BLOCK + 5 FLAG patterns).

Next.js | FastAPI | PostgreSQL

Silver 2025

AI-Powered Todo (H2)

89 tests passing. Full-stack Constitutional AI todo. 70% code reuse from H1.

Next.js | TypeScript | FastAPI

Silver 2024

Course Companion FTE (H1)

Zero-Backend-LLM architecture. Digital FTE model. 70% code reuse from H0.

Next.js | TypeScript | OpenAI

Bronze 2024

Personal AI CTO (H0)

Constitutional AI foundation. Established spec-first, four-session methodology.

TypeScript | Constitutional AI

Flagship Projects

The Autonomous Agent Ecosystem

Shipped

21 open-source platforms, one per problem domain

21 multi-agent platforms, each with its own FastAPI gateway, Helm chart, test suite and public repository — reference implementations running on synthetic domain data. 736 test functions across the 21 repositories, counted from source on 2026-09-10; 6 of them run those tests in CI. All open source.

FastAPI | Pytest | Helm

FinAgent-Nexus

Shipped

Multi-Agent Financial Intelligence

Settles 8 of 14 compliance principles by arithmetic — a full screen in under 2ms with zero model calls, and 94 tests that pass with no API key and no network.

LangGraph | Constitutional AI | Python 3.12

AI TradeFlow

Shipped

Inventory & Accounting for Wholesalers

Business-isolated books with a row-locked, idempotent posting path. 142 tests, including two-business leak tests and PostgreSQL oversell races. Credit-limit guard, void-by-reversal, WhatsApp collections desk, and Munshi AI with 5 read-only tools behind a deterministic constitution. Live no-signup demo; no production customer yet.

OpenAI Agents SDK | FastAPI | PostgreSQL

DevUnity Platform

Shipped

This portfolio — and its own proof

A triage orchestrator over four specialists, a real MCP server on the official SDK, and a written constitution verified blocking 4 of 4 violations with no model reachable.

Agents SDK | FastMCP | Next.js 15

Bazaar

Shipped

Unified B2B + B2C Marketplace

One codebase serving both retail checkout and a wholesale RFQ engine, with JazzCash, Easypaisa and card payments. PKR-native, three languages.

Next.js 15 | FastAPI | Supabase

Textile ERP Platform

Module 1 live

Multi-tenant fabric mill ERP, Module 1 of 7 live

Roll register, yarn stock ledger, weaving/knitting, LC landed cost and roll traceability, isolated by PostgreSQL row-level security. 42 tests against real Postgres; live no-signup demo. CMT billing, ledgers and financials are specced, not built.

FastAPI | PostgreSQL RLS | Next.js 15

Agent Factory (H5)

Shipped

A General Agent that manufactures Custom Agents

Two-tier architecture where Claude Code manufactures OpenAI Agents SDK agents; SKILL.md as portable intelligence units. Completed; deployed on Kubernetes + Dapr.

Claude Code | SKILL.md | Dapr

Key Projects

Textile ERP Platform (Pakistan)

In Development

2024 – Present

Founder

Full-scale ERP targeting Pakistan's textile and garment industry, from Fabric Mills to CMT units to garment exporters.

- Fabric Mill Module (complete, live demo): roll/lot register with partial issuance, append-only yarn stock ledger, weaving & knitting sessions, LC imports with landed cost, roll traceability, 7 deterministic Mill Pulse alerts; tenant isolation via PostgreSQL row-level security, 42 tests
- CMT Core (specced, next): order lifecycle, auto-billing (4 bill types), BOM inventory, production sessions, dispatch
- Financial (specced): party ledgers, aging, P&L and cash flow for multi-city hubs (Faisalabad, Sialkot, Gujranwala, Karachi, Lahore)

Next.js 15 | FastAPI | PostgreSQL RLS | SQLAlchemy 2 | Vercel

Live Demo

Agent Factory (H5)

Panaversity Hackathon 2025

2025

Lead Developer

Two-tier agentic architecture: General Agent manufactures Custom Agents at scale.

- General Agent (Claude Code) manufactures Custom Agent (OpenAI Agents SDK)
- SKILL.md files as portable, reusable, monetizable intelligence units
- Digital FTE pricing model targeting OpenAI Apps ecosystem (800M users)
- Deployed on Kubernetes + Dapr

Claude Code | OpenAI SDK | MCP | SKILL.md | Dapr | Kubernetes

Professional Experience

Founder & CEO, CMT Stitching Operations
Texcot Embroidery Sourcing House

2020 – Present
Karachi, Pakistan

- Engineered a multi-agent orchestration layer for financial-services research and Shari'ah screening, with separation of powers enforced structurally — no agent holds two of market data, weighting and verdict — rather than by prompt
- Implemented MCP interoperability so agent capability is exposed as read-only tools on a protocol server, verified against a live client, instead of as bespoke HTTP endpoints
- Architected deterministic guardrails from a written constitution, with a pattern layer that needs no model — verified blocking 4 of 4 violations with no model reachable, so enforcement survives a provider outage
- Architected the Textile ERP Platform — order lifecycle, four auto-billing types, party ledgers and BOM inventory — to digitize Pakistan's textile value chain
- Founded and operate a CMT stitching unit of 30–35 staff, running the full manufacturing lifecycle from sample fabric and design analysis through machine allocation, bulk production and inspection
- Lead the business's digital marketing across social channels and lead-generation funnels

Marketing Manager
JK Embroidery

2016 – 2020
Karachi, Pakistan

- Directed marketing and production planning for embroidery manufacturing operations
- Engineered staff and equipment scheduling to clear production bottlenecks, implementing process improvements that held under peak order flow
- Led a team of 30–35 staff, setting priorities against available resources and work-order flow

Country Manager / Buying Agent
Steven Berry

2003 – 2005
International

- Managed and organized international shipments, providing clients guidance throughout procurement processes
- Ensured products met established international quality standards across garment and textile categories
- Conducted pre-production, in-production, and final inspection of apparel, accessories, and textile products

Digital Marketing Services

Digital marketing strategy for Pakistani SMEs, and on offer to real estate and construction teams in the UAE:

- Social media management (Facebook, Instagram, X/Twitter, LinkedIn) + paid advertising campaigns
- Property portal listings, lead generation funnels, and monthly performance reporting
- Tailored strategies for GCC/expat/Pakistani diaspora investors; AED-priced service packages

Completed Courses

// CREDENTIALS

Certifications & Credentials

9 completed courses and certificates from Cognizant & Anthropic. Every card links to the issuer's own record or the certificate itself.

CERTIFICATE OF COMPLETION

Agent Foundations

Agent Academy · Cognizant AI Lab

Completed September 30, 2026

View certificate

VERIFIED COMPLETION BADGE

AI Fluency: Framework and foundations

Claude Academy · Anthropic's learning platform

Issued September 18, 2026

Verify with issuer

VERIFIED COMPLETION BADGE

Introduction to Claude Cowork

Claude Academy · Anthropic's learning platform

Issued September 18, 2026

Verify with issuer

VERIFIED COMPLETION BADGE

Claude Code 101

Claude Academy · Anthropic's learning platform

Issued September 18, 2026

Verify with issuer

VERIFIED COMPLETION BADGE

AI Fluency for builders

Claude Academy · Anthropic's learning platform

Issued September 18, 2026

Verify with issuer

VERIFIED COMPLETION BADGE

AI capabilities and limitations

Claude Academy · Anthropic's learning platform

Issued September 18, 2026

Verify with issuer

VERIFIED COMPLETION BADGE

Claude 101

Claude Academy · Anthropic's learning platform

VERIFIED COMPLETION BADGE

Claude Code in action

Claude Academy · Anthropic's learning platform

Issued September 18, 2026

Verify with issuer

Verify with issuer

VERIFIED COMPLETION BADGE

Introduction to Model Context Protocol

Claude Academy · Anthropic's learning platform

Verify with issuer

Education & Certifications

Governor Sindh Initiative for Artificial Intelligence & Computing (Panaversity) — Feb 2024 – Present
Advanced study in Agentic AI and Generative AI

Forward Deployed Engineer Training — Track B (Accelerated), 13 weeks: governed Vertical Systems of Record, stateless MCP (rev. 2026-07-28), multi-agent research systems, and structured extraction pipelines with validation and human review.

`TypeScript / JavaScript` `React / Next.js` `Python` `Agent SDKs & AI frameworks` `Loop / Harness / Graph architectures`

EXPECTED

Sindh Board / University of Karachi — Programme certification

Panaversity — PCAR-F — Architect Foundations, aligned to Anthropic's CCAR-F blueprint

Anthropic — Claude Certified Associate – Foundations → Architect – Foundations

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Panaversity Certified Associate: Foundations (PCAO-F) — Passed 25 Oct 2026
Panaversity

Alim (5-Year Islamic Studies Course)
Burooj Institute, Karachi

Associate Degree in Textile Technology — 1997
Textile Institute of Pakistan (APTMA)

H.S.C Pre-Engineering — 1994
Pakistan Shipowner's Government College, Karachi

S.S.C Science — 1992
Ladybird Grammar School, Karachi